# LAITH MUHAMMED

**Full-Stack Developer (Flutter • Next.js • Supabase) | Founder Mindset**

Baghdad, Iraq • Syrian National • Remote Available (UTC+3)

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Portfolio: laithmh/my portfolio

## PROFESSIONAL SUMMARY

Self-taught Full-Stack Developer with 4 years of dedicated product-building experience across cross-platform mobile apps, WebAssembly canvas tools, and production Next.js SaaS platforms. Founder of **Solen**, a turnkey Print-on-Demand marketplace engineered with 445 automated tests, and **B-Insta**, a multi-tenant link-in-bio platform built to self-fund operations. Skilled in architecting end-to-end systems from PostgreSQL relational schemas and Row-Level Security (RLS) to high-performance client UIs (Flutter BLoC & React 19). Complements strong technical execution with Google-certified e-commerce growth strategies and DaVinci Resolve motion video design.

## CORE COMPETENCIES & TECHNICAL SKILLS

- **Languages & Frameworks:** Dart, Flutter (Mobile & WASM), TypeScript, JavaScript, Next.js 16 (App Router), React 19, Python, HTML5, Tailwind CSS
- **Backend & Cloud:** Supabase, PostgreSQL, Row-Level Security (RLS), Serverless Edge Functions, RESTful APIs, FastAPI, Upstash Redis
- **Architecture & Systems:** BLoC Pattern, Clean Architecture, Reactive State Management, Android MediaCodec (H.264), WebSockets & P2P Sockets, Audio DSP (PCM Buffers)
- **Quality & DevOps:** Vitest (445 Unit/Integration Tests), Playwright (E2E), Git / GitHub, Vercel Deployment
- **Design & Creative Media:** DaVinci Resolve (Color Grading & Motion Graphics), Cinematic Mobile Filming, Figma UI/UX & Design Systems
- **E-Commerce & Growth:** Google Certified in Digital Marketing & E-Commerce, Localized Cash-on-Delivery (COD) Workflows, Courier Logistics Automation

## VENTURES & SOFTWARE PROJECTS

### Dukkani (دكّاني) — Principal Mobile Architect & Flutter Engineer *(Client Project in Syria • Actively Used in Production)*

*2026 | Latakia / Damascus, Syria*

*Commercial Retail POS & USD-Pegged Customer Debt Ledger*

- Architected and shipped a 100% offline-first Point of Sale & Customer Debt Ledger in **Flutter 3** & **Drift (SQLite WAL)** for a retail grocery merchant in Syria, replacing fragile paper debt notebooks with an inflation-protected append-only accounting system actively used in daily store operations under power and internet blackouts.
- Engineered an arbitrary-precision multi-currency engine using **Decimal Math** and custom SQLite converters, eliminating 100% of floating-point rounding errors and implementing a 50-step cyclic algorithm snapping checkout cash totals to physically payable Syrian banknote denominations (10, 25, 50 SYP) while preserving

exact line-item precision.
- Enforced financial data integrity with **4 native SQLite database triggers**, preventing unauthorized updates or deletions on issued invoices and ledger records while maintaining immutable audit trails with double-entry compensating reversals.
- Designed a zero-cloud offline licensing system using **asymmetric Ed25519 digital signatures and AES-256-GCM**, binding commercial activation to hardware device fingerprints with anti-clock-rollback tamper verification that cross-checks system time against SQLite transaction high-water marks.
- Created the "Salt & Pepper" elderly-friendly design system, establishing a WCAG AAA compliant monochrome interface (14.1:1 contrast) with a mandatory 56px/64px touch target floor, RTL Arabic layout with Western numeral normalization, reducing rush-hour customer checkout time to under 20 seconds.
- Authored **44 automated test suites** across 4 layers, verifying financial rounding boundaries, in-memory SQLite transactions (`NativeDatabase.memory()`), **flutter_bloc** state transitions, and responsive layouts from 320px devices to tablets.

### Solen (سولين) — Founder & Full-Stack Developer *(Active Startup • Pre-Launch)*

*June 2024 – Present | Baghdad, Iraq*

*Turnkey B2B2C Print-on-Demand & Creator Marketplace for Iraq*

- Architected the complete web platform using **Next.js 16 (App Router)**, **React 19**, **TypeScript**, and **Tailwind CSS**, delivering trilingual support (Arabic, Kurdish, English) with instant page loads.
- Engineered an in-browser 2D garment customizer using **Flutter compiled to WebAssembly (WASM)**, achieving 60 FPS canvas manipulation, real-time DPI resolution gauges, and boundary-safe printing constraints.
- Built a server-side graphics processing pipeline using **Node.js Sharp** to composite and export production-ready, 300 DPI transparent DTF print sheets directly to print houses.
- Designed a secure multi-tenant **Supabase & PostgreSQL** relational schema with 10+ core tables and strict **Row-Level Security (RLS)** protecting merchant earnings, catalog designs, and customer orders.
- Solved regional commerce hurdles by designing zero-float financial clearance and localized Cash on Delivery (COD) logistics covering all 18 Iraqi governorates, verified by **445 automated Vitest unit tests and Playwright E2E suites**.

### B-Insta — Founder & Full-Stack SaaS Developer *(Venture Project • Pre-Launch)*

*January 2024 – 2025 | Baghdad, Iraq*

*Multi-Tenant Social Commerce Link-in-Bio Storefront Platform*

- Engineered a multi-tenant SaaS platform empowering Instagram sellers to deploy mobile-first storefronts (`/s/[slug]`) with instant local cart caching and frictionless COD checkout.
- Automated local logistics by implementing governorate shipping rate calculation, courier CSV formatting (for Iraqi dispatchers), and WhatsApp soft confirmation flows.
- Implemented **Upstash Redis** sliding-window rate limiters to defend order endpoints against checkout abuse without impacting authentic customer checkout speed.
- Built with **Next.js 16 Server Actions**, **TypeScript**, **Supabase PostgreSQL**, and validated schemas with **Zod**.

### OffCast — Independent Systems & Mobile Developer *(Open Source Project)*

*2023 – 2024 | Baghdad, Iraq*

*Ultra-Low Latency Wireless Field Monitor & Smart Teleprompter*

- Built an offline peer-to-peer screen sharing system to solve personal framing and monitoring challenges during solo camera video shoots.
- Achieved sub-20ms glass-to-glass latency over direct Wi-Fi Hotspots without internet using **Flutter BLoC**, native **Kotlin UDP relay**, and **Android MediaCodec (H.264)** hardware streaming.
- Built an on-device smart teleprompter that automatically scrolls speech text based on voice detection using custom **PCM-16 RMS Audio DSP**.

### MusicLOTM — Independent Mobile Developer *(First Flutter Project • Daily Driver)*

*2022 – 2023 | Latakia / Baghdad*

*Offline-First Android Music Player & Native Audio Visualizer*

- Developed a complete Android music player from scratch as a foundational project to master Flutter and Dart.
- Engineered a 60 FPS real-time audio visualizer by tapping raw **ExoPlayer PCM byte buffers** in native Android, avoiding microphone permission requirements.
- Implemented in-app ID3 tag editing, persistent playlist management using **Hive NoSQL**, and a custom neumorphic UI across 11 screens.

## EDUCATION & CERTIFICATIONS

### Teshreen University — Latakia, Syria

*Undergraduate Coursework in Computer Science | 2021 – 2023*

- Completed 2 years of intensive university coursework in computer science fundamentals, algorithms, data structures, and software principles.
- Academic trajectory interrupted due to regional circumstances; transitioned immediately into disciplined, full-time independent software engineering and product building.

### Google via Coursera — Professional Certificate

*Google Digital Marketing & E-Commerce Professional Certificate | Completed 2026*

- Comprehensive 7-course credential covering digital marketing, conversion rate optimization (CRO), search engine optimization (SEO), customer acquisition, and e-commerce analytics.

## LANGUAGES

- **Arabic:** Native

**English:** Professional Working Proficiency (Fluent Reading & Writing, Intermediate Speaking)